

9-xalqaro fizika olimpiadasining masalalari

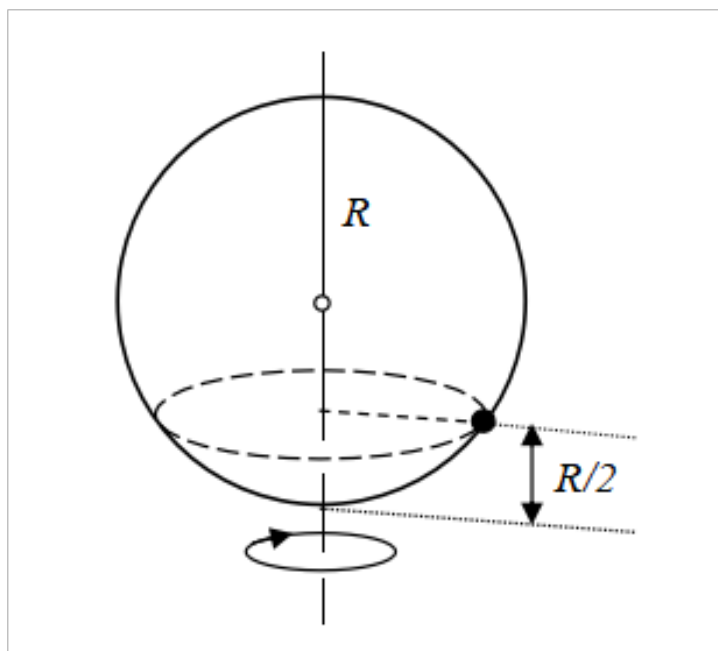
(Budapesht, Vengriya, 1976)

Nazariy masalalar

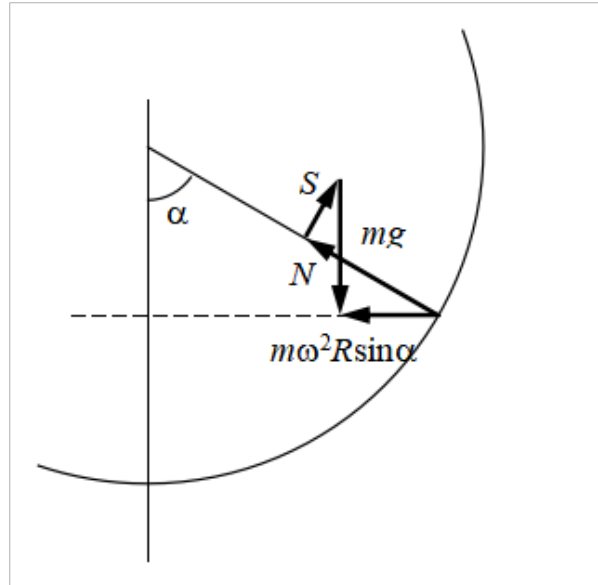
1-masala

Radiusi $R = 0.5$ m bo'lgan ichi bo'sh sfera o'z markazidan o'tuvchi vertikal o'q atrofida $\omega = 5 \text{ s}^{-1}$ burchak tezlik bilan aylanadi. Sfera ichida kichik bir blok sfera bilan birgalikda $R/2$ balandlikda harakatlanmoqda (6-rasm). ($g = 10 \text{ m/s}^2$.)

- a) Bu shartni qanoatlantirish uchun ishqalanish koeffitsiyenti kamida qancha bo'lishi kerak?
- b) $\omega = 8 \text{ s}^{-1}$ hol uchun ham minimal ishqalanish koeffitsiyentini toping.
- c) Ikkala holatda ham turg'unlik masalasini tekshiring:
 - α) blok holatining kichik o'zgarishi uchun,
 - β) sfera burchak tezligining kichik o'zgarishi uchun.



6-rasm



7-rasm

Yechim

a) Blok radiusi $R \sin \alpha$ bo'lgan gorizontaal aylana bo'ylab harakatlanadi. Blokka ta'sir qiluvchi natijaviy kuch shu aylana markaziga yo'nalgan (7-rasm). Devor tomonidan ta'sir qiluvchi normal kuch N , ishqalanish kuchi S va og'irlik kuchi mg ning vektor yig'indisi natijaviy kuchga teng: $m\omega^2 R \sin \alpha$.

Gorizontaal va vertikal tashkil etuvchilar orasidagi bog'lanishlar:

$$\begin{aligned} m\omega^2 R \sin \alpha &= N \sin \alpha - S \cos \alpha, \\ mg &= N \cos \alpha + S \sin \alpha. \end{aligned}$$

Tenglamalar sistemasining yechimi:

$$\begin{aligned} S &= mg \sin \alpha \left(1 - \frac{\omega^2 R \cos \alpha}{g} \right), \\ N &= mg \left(\cos \alpha + \frac{\omega^2 R \sin^2 \alpha}{g} \right). \end{aligned}$$

Blok pastga sirpanib ketmaydi, agar

$$\mu_a \geq \frac{S}{N} = \sin \alpha \cdot \frac{1 - \frac{\omega^2 R \cos \alpha}{g}}{\cos \alpha + \frac{\omega^2 R \sin^2 \alpha}{g}} = \frac{3\sqrt{3}}{23} \approx 0.2259.$$

Bu holda sirpanishni, ya'ni pastga siljishni oldini olish uchun kamida shu ishqalanish bo'lishi kerak.

b) Agar aksincha $\frac{\omega^2 R \cos \alpha}{g} > 1$ bo'lsa, blokning yuqoriga sirpanib ketishini oldini olish uchun biroz ishqalanish zarur. $m\omega^2 R \sin \alpha$ kuch S , N va mg kuchlarining natijaviysiga

teng bo'lishi kerak. Minimal ishqalanish ko'effitsiyenti sharti (8-rasm):

$$\mu_b \geq \frac{S}{N} = \sin \alpha \cdot \frac{\frac{\omega^2 R \cos \alpha}{g} - 1}{\cos \alpha + \frac{\omega^2 R \sin^2 \alpha}{g}} = \frac{3\sqrt{3}}{29} \approx 0.1792.$$

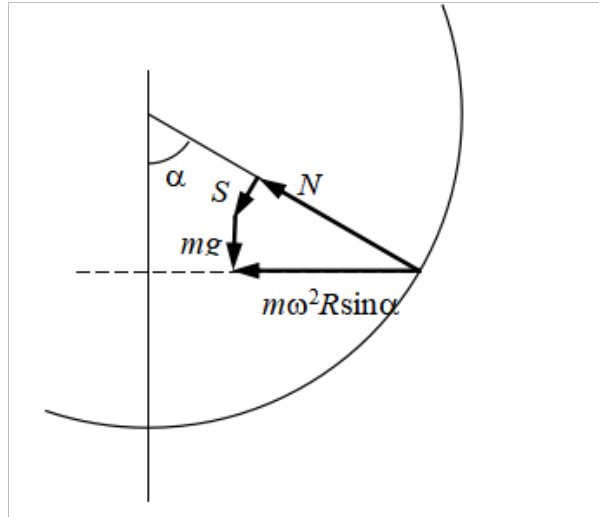


Figure 1: 8-rasm

c) a) va b) holatlar uchun μ_a va μ_b ni α va ω funksiyalari sifatida tekshirishimiz kerak (9/a-rasm va 9/b-rasmga qarang):

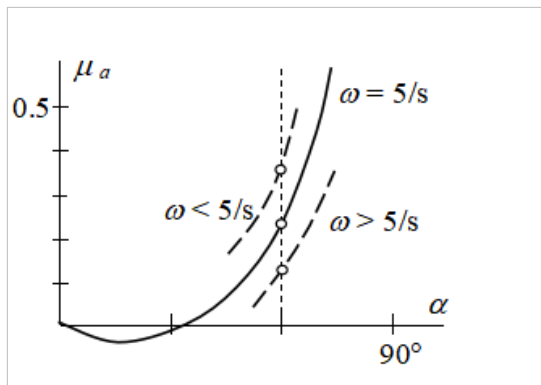


Figure 2: 9a rasm

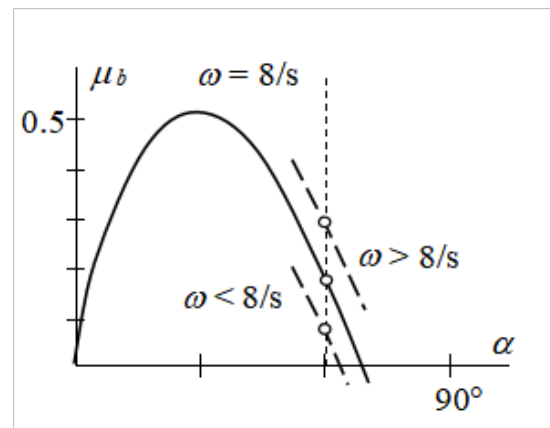


Figure 3: 9b rasm

a) holda: agar blok yuqoriga sirpansa, u orqaga qaytadi; agar pastga sirpansa, qaytib kelmaydi. Agar ω ortsa, blok muvozanatda qoladi, agar ω kamaysa, blok pastga sirpanadi.

b) holda: agar blok yuqoriga sirpansa, u shu yerda qoladi; agar blok pastga sirpansa, orqaga qaytadi. Agar ω ortsa, blok yuqoriga ko'tariladi, agar ω kamaysa, blok muvozanatda qoladi.

2-masala

Asosining yuzi 1 dm^2 bo'lgan silindrning devorlari, porsheni va ichki ajratuvchi devori mukammal issiqlik izolyatorlaridir (*10-rasm*). Ajratuvchi devordagi klapan, agar o'ng tomondagi bosim chap tomondagidan katta bo'lsa ochiladi. Dastlab chap tomonda 12 g geliy, o'ng tomonda 2 g geliy bor. Ikkala tomonning uzunligi 11.2 dm dan va harorat 0°C . Tashqarida bosim 100 kPa . O'zgarmas hajmdagi solishtirma issiqlik sig'imi $c_v = 3.15 \text{ J/gK}$, o'zgarmas bosimdagi solishtirma issiqlik sig'imi $c_p = 5.25 \text{ J/gK}$. Porshen asta-sekin ajratuvchi devor tomon suriladi. Klapan ochilganda to'xtatiladi, so'ngra devorga yetguncha yana sekin surish davom ettiriladi. Porshen ustida biz bajargan ishni toping.

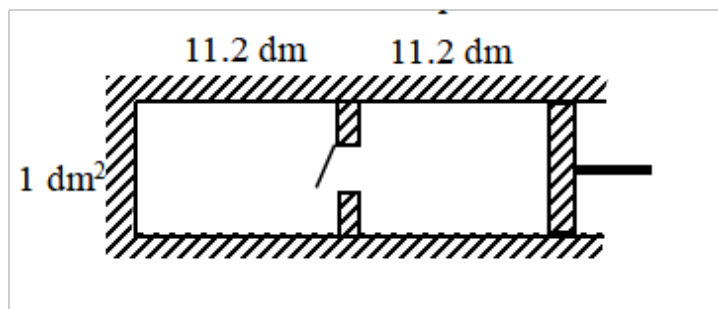


Figure 4: 10-rasm

Yechim

0°C harorat va 100 kPa bosimda 4 g geliyning hajmi 22.4 dm^3 (molyar hajm). Bundan kelib chiqadiki, dastlab chap tomondagi bosim 600 kPa , o'ng tomondagi bosim 100 kPa . Demak, klapan yopiq.

O'ng tomondagi bosim 600 kPa ga yetguncha adiabatik siqilish sodir bo'ladi ($\kappa = 5/3$):

$$100 \cdot 11.2^{5/3} = 600 \cdot V^{5/3},$$

bundan o'ng tomondagi hajm (klapan ochilganda):

$$V = 3.82 \text{ dm}^3.$$

Ideal gaz tenglamasidan bu nuqtada o'ng tomondagi harorat:

$$T_1 = \frac{pV}{nR} = 552 \text{ K}.$$

Bu fazada bajarilgan barcha ish gazning ichki energiyasini oshiradi:

$$W_1 = (3.15 \text{ J/gK}) \cdot (2 \text{ g}) \cdot (552 \text{ K} - 273 \text{ K}) = 1760 \text{ J}.$$

Keyin klapan ochiladi, porshen to'xtatiladi. Aralashish tugagandan keyingi harorat:

$$T_2 = \frac{12 \cdot 273 + 2 \cdot 552}{14} = 313 \text{ K.}$$

Bu fazada energiya o'zgarmaydi, porshen ustida ish bajarilmaydi.

So'ngra $11.2 + 3.82 = 15.02 \text{ dm}^3$ dan 11.2 dm^3 gacha adiabatik siqilish sodir bo'ladi:

$$313 \cdot 15.02^{2/3} = T_3 \cdot 11.2^{2/3},$$

bundan

$$T_3 = 381 \text{ K.}$$

Bajarilgan barcha ish gazning energiyasini oshiradi:

$$W_3 = (3.15 \text{ J/gK}) \cdot (14 \text{ g}) \cdot (381 \text{ K} - 313 \text{ K}) = 3000 \text{ J.}$$

Bajarilgan to'liq ish:

$$W_{\text{total}} = W_1 + W_3 = 4760 \text{ J.}$$

Tashqi atmosfera bosimi bajargan ishni ayirish kerak:

$$W_{\text{atm}} = 100 \text{ kPa} \cdot 11.2 \text{ dm}^3 = 1120 \text{ J.}$$

Bizning porshen ustida bajargan ishimiz:

$$W = W_{\text{total}} - W_{\text{atm}} = \mathbf{3640 \text{ J.}}$$

3-masala

Shisha sferaning ichida biror joyda havo pufakchasi bor. Sferaga zarar yetkazmagan holda pufakcha diametrini aniqlash usullarini tavsiflang.

Yechim

Biz shisha zichligining biron-bir qiymatiga tayanib bo'lmaydi. Bu juda noaniq. Sindirish ko'rsatkichini pufakchaga tegmaydigan yorug'lik dastasi yordamida aniqlash mumkin. Boshqa bir usul sferani xuddi shunday sindirish ko'rsatkichiga ega suyuqlikka botirishdan iborat: uning sirti ko'rinmas bo'lib qoladi.

Ko'plab usullarni topish mumkin.

Sfera va pufakcha markazlarini birlashtiruvchi chiziq – o'qni aniqlashdan boshlashimiz mumkin. Eng oson yo'li – «ag'darish» usulidan foydalanish. Agar sfera gorizontal tekislikka qo'yilsa, o'q vertikal holatni egallaydi. Pufakchaning o'q bo'ylab ikkala yo'nalishdan ko'ringan tasviri aylanadir.

Agar sfera sindirish ko'rsatkichi bir xil bo'lgan suyuqlikka botirilsa, sferik pufakcha amalda parallel plastinka ichida bo'ladi (11-rasm). Uning chegaralarini mikrometr yordamida yoki parallel yorug'lik dastalari yordamida aniqlash mumkin.

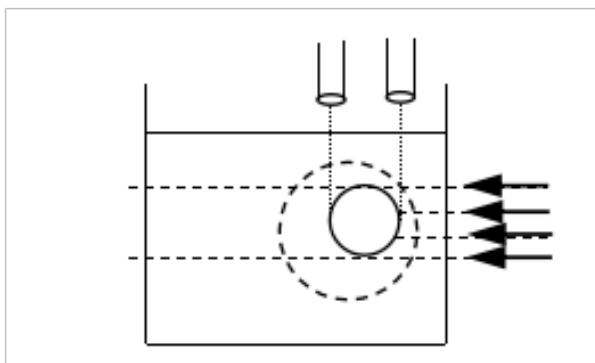


Figure 5: 11-rasm

O'q bo'ylab biz ikkita qalin manfiy linzadan iborat linzalar sistemasiga ega bo'lamiz. Pufakcha diametrini bir nechta o'lchashlar va murakkab hisob-kitoblar orqali aniqlash mumkin.

Agar shishaning sindirish ko'rsatkichi ma'lum bo'lsa, biz o'qning oxirida sferaga xuddi shunday sindirish ko'rsatkichiga ega bo'lgan plano-botiq linzani moslashtirishimiz mumkin (12-rasm). ABCD parallel plastinka hosil qilgani uchun pufakcha diametrini parallel yorug'lik dastalari yordamida o'lchash mumkin.

Yorug'lik dastasini sfera sirtining A nuqtasiga fokuslab (13-rasm), sfera ichida A nuqtadan tarqaluvchi dasta hosil qilamiz. Nurlar boshqa tomondagi sirtga urilib, biror qalpoqchani yoritadi. Sferik qalpoqchani o'lchab, ϕ burchakni olamiz. ψ burchakni xuddi

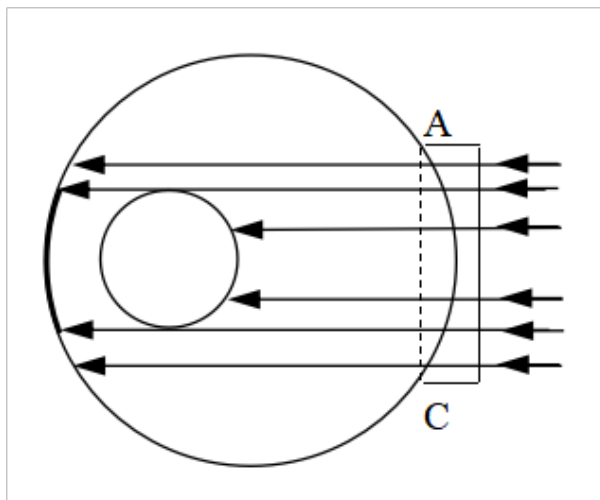


Figure 6: 12-rasm

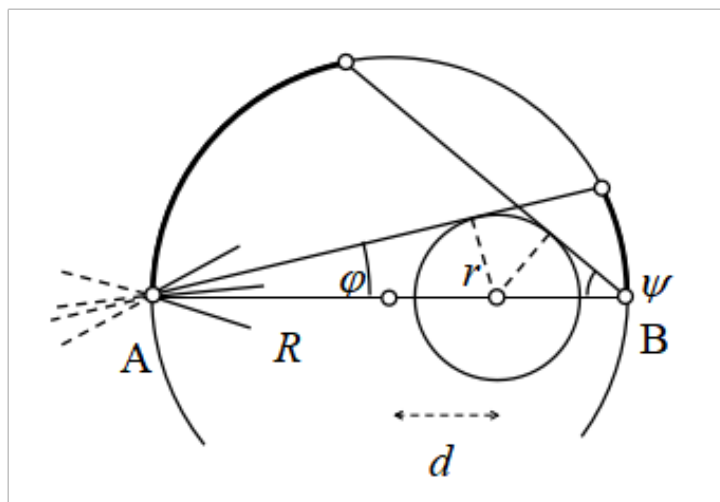


Figure 7: 13-rasm

shunday B nuqtada olish mumkin. Quyidagilardan:

$$\sin \phi = \frac{r}{R + d} \quad \text{va} \quad \sin \psi = \frac{r}{R - d}$$

quyidagilarga ega bo'lamiz:

$$r = 2R \cdot \frac{\sin \psi \sin \phi}{\sin \psi + \sin \phi}, \quad d = R \cdot \frac{\sin \psi - \sin \phi}{\sin \psi + \sin \phi}.$$

Pufakcha diametrini rentgen nurlari yordamida ham aniqlash mumkin. Rentgen nurlari shishada sinmaydi. Ular jismning tuzilishini, bizning holatda pufakchanning joylashuvi va diametrini ko'rsatuvchi soyalarni tushiradi.

Shuningdek, o'qqa nisbatan inersiya momentini va shu orqali pufakcha diametrini aniqlashimiz mumkin.

Eksperimental masala

O'quvchilarga berilgan to'liq matn:

Ish joyida boshqa asboblardan qatorida 12 V li elektr isitgichli probirka, solishtirma issiqlik sig'imi ma'lum suyuqlik ($c_0 = 2.1 \text{ J/g}^\circ\text{C}$) va issiqlik xossalari noma'lum bo'lgan X moddasi mavjud. X moddasi suyuqlikda erimaydi.

X kristall moddasining issiqlik xossalarini xona harorati va 70°C orasida tekshiring. X moddasining issiqlik parametrlarini aniqlang. O'lchangan ma'lumotlarni jadvalga kiriting va grafikda chizing.

(Faqat stolda tayyorlab qo'yilgan asbob va materiallardan foydalanishingiz mumkin. Shikastlangan asboblardan va ishlatib bo'lingan materiallar almashtirilmaydi.)

Yechim

Avval suyuqlikni, so'ngra suyuqlik va kristall moddani birgalikda qizdirib, ikkita vaqt-harorat grafigini chizish mumkin. Grafiklardan solishtirma issiqlik sig'imi, erish nuqtasi va erish issiqligini osongina olish mumkin.

Adabiyotlar

1. W. Gorzkowski: Problems of the 1st International Physics Olympiad, *Physics Competitions* **5**, no2 pp6-17, 2003.
2. R. Kunfalvi: *Collection of Competition Tasks from the 1st through XVth International Physics Olympiads 1967-1984*, Roland Eötvös Physical Society in cooperation with UNESCO, Budapest, 1985.
3. *A Nemzetközi Fizikai Diákolimpiák feladatai I.-XV.*, Eötvös Loránd Fizikai Társulat, Középiskolai Matematikai Lapok, 1985.